

1. Linear Algebra

Cauchy–Schwarz: (special case of Hölder)

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$$

Hölder:

$$|\langle x, y \rangle| \leq \|x\|_p \|y\|_{p'}, \quad \frac{1}{p} + \frac{1}{p'} = 1$$

Minkowski's inequality:

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

Matrix basics:

$$\bullet \operatorname{tr}(AB) = \operatorname{tr}(BA) \quad \bullet \langle A, B \rangle_F = \operatorname{tr}(A^\top B)$$

Trace and determinant:

$$\operatorname{trace}(A) = \sum_i \lambda_i \quad \det(A) = \prod_i \lambda_i$$

Extreme eigenvalues:

$$\lambda_1 = \max_{\|x\|_2=1} x^\top A x \quad \lambda_p = \min_{\|x\|_2=1} x^\top A x$$

PSD through eigenvalues: $A \in \mathbb{S}^p$

$$A \succeq 0 \iff \lambda_i(A) \geq 0 \quad \forall i$$

Induced operator norm: $A \in \mathbb{R}^{m \times n}$

$$\|A\|_{p \rightarrow q} = \sup_{\|x\|_p \leq 1} \|Ax\|_q = \sup_{x \neq 0} \frac{\|Ax\|_q}{\|x\|_p}$$

Composition bound (Submultiplicativity):

$$\|AB\|_{p \rightarrow r} \leq \|A\|_{q \rightarrow r} \|B\|_{p \rightarrow q}$$

Spectral norm: $p = q = 2$

$$\|A\|_{\text{op}} = \|A\|_{2 \rightarrow 2} = \sup_{\|x\|_2=1} \|Ax\|_2 = \sigma_1(A) = \sqrt{\lambda_1(A^\top A)}$$

Frobenius norm:

$$\|A\|_F^2 = \sum_{i,j} A_{ij}^2 = \operatorname{tr}(A^\top A) = \sum_{i=1}^{\operatorname{rank}(A)} \sigma_i(A)^2$$

Spectral vs Frobenius norm:

$$\|A\|_{\text{op}} \leq \|A\|_F \leq \sqrt{\operatorname{rank}(A)} \|A\|_{\text{op}}$$

Frobenius submultiplicativity: A, B conformable matrices,

$$\|AB\|_F \leq \|A\|_{\text{op}} \|B\|_F \leq \|A\|_F \|B\|_F$$

Rank one case: if $A = uv^\top$ then

$$\|A\|_{\text{op}} = \|A\|_F = \|u\|_2 \|v\|_2$$

2. Probability Basics

Variance:

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y)$$

L_p -norms:

$$\|X\|_{L_p} = (\mathbb{E}|X|^p)^{1/p}, \quad \|X\|_{L_\infty} = \operatorname{ess\,sup}|X|$$

Key inequalities

Hölder: p, p' are conjugate exponents,

$$|\mathbb{E}[XY]| \leq \|X\|_{L_p} \|Y\|_{L_{p'}}$$

Minkowski:

$$\|X + Y\|_{L_p} \leq \|X\|_{L_p} + \|Y\|_{L_p}$$

Jensen (convex ϕ):

$$\phi(\mathbb{E}X) \leq \mathbb{E}[\phi(X)]$$

Since **norms are convex**, $\|\mathbb{E}X\| \leq \mathbb{E}\|X\|$

Monotonicity of L_p -norms: for $1 \leq p \leq q \leq \infty$,

$$\|X\|_{L_p} \leq \|X\|_{L_q}$$

L_p vs ℓ_p norms: Let $x = (x_1, \dots, x_n)$ and X be a random variable that takes values x_i each with prob. $1/n$ then,

$$\|X\|_{L_p} \leq 1 \iff \|x\|_{\ell_p} \leq n^{1/p}$$

Integrated tail identities:

$$\mathbb{E}|X|^p = \int_0^\infty p t^{p-1} \mathbb{P}\{|X| > t\} dt$$

For a non-decreasing $\Phi \geq 0$ with $\Phi(0) < \infty$,

$$\mathbb{E}[\Phi(|X|)] = \Phi(0) + \int_0^\infty \Phi'(t) \mathbb{P}\{|X| > t\} dt$$

Moment growth bound: Random variable X , assume $\forall t \geq 0$,

$$\mathbb{P}\{|X| \geq t\} \leq C \exp(-ct^\alpha),$$

then there exists a constant $C' = C'(C, \alpha)$ such that

$$\|X\|_{L_p} = (\mathbb{E}|X|^p)^{1/p} \leq C'(p/c)^{1/\alpha}, \quad \text{for all } p \geq 1.$$

Markov: if $Y \geq 0$, then for $t > 0$,

$$\mathbb{P}\{Y \geq t\} \leq \frac{\mathbb{E}[Y]}{t}.$$

Chebyshev:

$$\mathbb{P}\{|X - \mathbb{E}X| \geq t\} \leq \frac{\operatorname{Var}(X)}{t^2}.$$

Cantelli (one-sided Chebyshev):

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq \frac{\operatorname{Var}(X)}{\operatorname{Var}(X) + t^2}.$$

Paley–Zygmund. $X \geq 0$ and $\mathbb{E}X^2 < \infty$, then $\forall \lambda \in (0, 1)$,

$$\mathbb{P}\{X \geq \lambda \mathbb{E}X\} \geq (1 - \lambda^2) \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2}$$

3. Exponential Concentration

Polynomial moment method:

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq \inf_{p \in \mathbb{N}} \frac{\mathbb{E}[(X - \mathbb{E}X)_+^p]}{t^p}$$

Exponential moment method (Chernoff bound):

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq \inf_{\lambda > 0} \exp(-\lambda t + \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)})$$

4. MGF and CGF

Moment Generating Function (MGF):

$$m_X(\lambda) = \mathbb{E}e^{\lambda X} \in (0, \infty] \quad (\lambda \in \mathbb{R})$$

Cumulant generating function (CGF):

$$\kappa_X(\lambda) = \log \mathbb{E}e^{\lambda X} \in (-\infty, \infty] \quad (\lambda \in \mathbb{R})$$

Centered CGF:

$$\psi_X(\lambda) = \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)} = \kappa_X(\lambda) - \lambda \mathbb{E}X$$

Moments from MGF:

$$m_X(\lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \mathbb{E}[X^k] \implies m_X^{(k)}(0) = \mathbb{E}[X^k]$$

Convexity of CGF: If $m_X(\lambda) < \infty$ in a neighborhood of a point λ , then κ_X is twice differentiable there and

$$\kappa_X''(\lambda) = \operatorname{Var}_{\mathbb{P}_\lambda}(X) \geq 0,$$

where $\mathbb{P}_\lambda$ is the tilted measure

$$d\mathbb{P}_\lambda = \frac{e^{\lambda X}}{\mathbb{E}[e^{\lambda X}]} d\mathbb{P}$$

Useful result near 0: if m_X is finite near 0,

$$\psi_X(0) = 0, \quad \psi_X'(0) = 0, \quad \psi_X''(0) = \operatorname{Var}(X)$$

5. Sub-Gaussian

Definition: X is σ^2 -sub-Gaussian if

$$\psi_X(\lambda) = \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)} \leq \frac{\sigma^2 \lambda^2}{2} \quad \forall \lambda \in \mathbb{R}$$

ψ_2 norm: X is sub-Gaussian iff

$$\|X\|_{\psi_2} = \inf\{K > 0 : \mathbb{E} \exp\left(\frac{X^2}{K^2}\right) \leq 2\} < \infty$$

Equivalent characterizations: Define $Z = X - \mathbb{E}X$
Gaussian tails:

$$P(|Z| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right), \quad t \geq 0$$

Moment growth:

$$\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p} \leq C \sigma \sqrt{p}, \quad p \geq 1$$

Example:

$$\bullet X \sim \mathcal{N}(0, \sigma^2) \implies X \text{ is } \sigma^2\text{-sub-Gaussian w/ } \|X\|_{\psi_2} \asymp \sigma$$

6. Hoeffding Bounds

Hoeffding's lemma: if $X \in [a, b]$ (a.s.), then for all $\lambda \in \mathbb{R}$,

$$\psi_X(\lambda) = \log \mathbb{E} e^{\lambda(X - \mathbb{E}X)} \leq \frac{\lambda^2(b-a)^2}{8}$$

Hoeffding's inequality: (X_i) are independent w/ $X_i \in [a_i, b_i]$ (a.s.) $\forall i$. Let $S_n = \sum_i X_i$ and define the **variance proxy**:

$$v = \frac{1}{4} \sum_{i=1}^n (b_i - a_i)^2,$$

then for all $t \geq 0$,

$$\mathbb{P}\{|S_n - \mathbb{E}S_n| \geq t\} \leq 2 \exp\left(-\frac{t^2}{2v}\right)$$

Sub-gaussian Hoeffding inequality: (X_i) are indep, mean-zero, and sub-Gaussian w/ $K_i = \|X_i\|_{\psi_2}$. Then, for every $t \geq 0$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \geq t\right\} \leq 2 \exp\left(-c \frac{t^2}{\sum_i K_i^2}\right)$$

Equivalently: $\|\sum_i X_i\|_{\psi_2}^2 \leq C \sum_i \|X_i\|_{\psi_2}^2$

7. Sub-Exponential

Definition: X is sub-exponential if

$$\|X\|_{\psi_1} = \inf\{K > 0 : \mathbb{E} \exp\left(\frac{|X|}{K}\right) \leq 2\} < \infty$$

Equivalent characterizations: Define $Z = X - \mathbb{E}X$

Exponential tails:

$$\mathbb{P}(|Z| \geq t) \leq 2 \exp(-c \frac{t}{K}), \quad t \geq 0$$

Local CGF Control:

$$\psi_Z(\lambda) = \log \mathbb{E} e^{\lambda(X - \mathbb{E}X)} \leq CK^2 \lambda^2 \quad |\lambda| \leq \frac{c}{K}$$

Moment growth:

$$\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p} \leq CKp, \quad p \geq 1$$

Closure Properties:

- X is sub-Gaussian, X^2 sub-exponential, $\|X^2\|_{\psi_1} \asymp \|X\|_{\psi_2}^2$
- X, Y is sub-Gaussian, then $\|XY\|_{\psi_1} \lesssim \|X\|_{\psi_2} \|Y\|_{\psi_2}$

8. Bernstein inequality

(X_i) are independent, mean-zero sub-exponential. Set

$$V = \sum_{i=1}^n \|X_i\|_{\psi_1}^2, \quad B = \max_i \|X_i\|_{\psi_1}$$

Then for all $t \geq 0$,

$$\mathbb{P}\left\{\left|\sum_{i=1}^n X_i\right| \geq t\right\} \leq 2 \exp\left[-c \min\left(\frac{t^2}{V}, \frac{t}{B}\right)\right]$$

Structure in L_p bound: $\|\sum_i X_i\|_{L_p} \lesssim \sqrt{p} \sqrt{V} + pB$

9. Metric Entropy

Throughout, let (T, d) be a metric space, let $K \subset T$, and let $\epsilon > 0$.

ϵ -net A set $\mathcal{M} \subset K$ is an ϵ -net of K if for every $x \in K$ there exists $y \in \mathcal{M}$ such that

$$d(x, y) \leq \epsilon.$$

Equivalently, the balls of radius ϵ centered at points of $\mathcal{M}$ cover K .

covering number The *covering number* of K is

$$\mathcal{N}(K, d, \epsilon) := \min\{|\mathcal{M}| : \mathcal{M} \subset K \text{ is an } \epsilon\text{-net of } K\}.$$

ϵ -separated A set $\mathcal{M} \subset K$ is ϵ -separated if

$$d(x, y) > \epsilon \quad \text{for all distinct } x, y \in \mathcal{M}.$$

packing number The *packing number* of K is

$$\mathcal{P}(K, d, \epsilon) := \max\{|\mathcal{M}| : \mathcal{M} \subset K \text{ is } \epsilon\text{-separated}\}.$$

metric entropy The *metric entropy* is the logarithm of the covering number:

$$H(K, d, \epsilon) := \log \mathcal{N}(K, d, \epsilon).$$

10. Covariance Estimation

The ϵ -net reduction

Lemma (Net reduction for operator norms). Let A be an $m \times n$ matrix and let $\mathcal{N}$ be an ϵ -net of S^{n-1} in $\|\cdot\|_2$ with $\epsilon \in (0, 1)$. Then

$$\|A\| = \sup_{\|x\|_2=1} \|Ax\|_2 \leq \frac{1}{1-\epsilon} \sup_{x \in \mathcal{N}} \|Ax\|_2.$$

Lemma (Net reduction for symmetric matrices). Let $A \in \mathbb{R}^{d \times d}$ be symmetric and let $\mathcal{N}$ be an ϵ -net of S^{d-1} with $\epsilon \in (0, 1/2)$. Then

$$\|A\| = \sup_{\|x\|_2=1} |x^\top A x| \leq \frac{1}{1-2\epsilon} \sup_{x \in \mathcal{N}} |x^\top A x|.$$

Perturbation theory

Lemma (Eigenvalues). For symmetric $d \times d$ matrices A, B ,

$$\max_{1 \leq i \leq d} |\lambda_i(A) - \lambda_i(B)| \leq \|A - B\|.$$

Thus, once we have an operator-norm bound on $\widehat{\Sigma} - \Sigma$, all eigenvalues are uniformly close.

Eigenvectors are only stable when there is a gap separating the relevant part of the spectrum.

Theorem 4.2 (Davis-Kahan for top- k subspaces). Let A, B be symmetric $d \times d$ matrices. Let P_A be the orthogonal projector onto $\text{span}\{v_1(A), \dots, v_k(A)\}$. Assume there is an eigengap at k for A ,

$$\delta := \lambda_k(A) - \lambda_{k+1}(A) > 0,$$

and that $\|A - B\| \leq \delta/2$. Let P_B be the projector onto $\text{span}\{v_1(B), \dots, v_k(B)\}$. Then

$$\|P_A - P_B\| \leq \frac{2\|A - B\|}{\delta}.$$

11. Random Process Theory

A random process is simply a family of random variables indexed by a set.

Definition (Random process). Let T be a nonempty set. A *random process* indexed by T is a family of real-valued random variables

$$(X_t)_{t \in T}.$$

Main Goal The central quantity we will study is the expected supremum

$$\mathbb{E} \sup_{t \in T} X_t.$$

This quantity measures the typical size of the largest fluctuation of the process over its index set.

Lemma (Finite maximal inequality). Let $Z_1, \dots, Z_M$ be centered random variables such that

$$\|Z_j\|_{\psi_2} \leq L \quad \text{for all } j = 1, \dots, M.$$

Then

$$\mathbb{E} \max_{1 \leq j \leq M} Z_j \leq CL \sqrt{\log M},$$

where $C > 0$ is an absolute constant.

Theorem (Dudley's entropy integral inequality). Let $(X_t)_{t \in T}$ be a centered random process on a metric space (T, d) such that

$$\|X_t - X_s\|_{\psi_2} \leq K d(t, s) \quad \text{for all } s, t \in T.$$

Assume T is finite, then

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \int_0^{\text{diam}(T, d)} \sqrt{\log \mathcal{N}(T, d, \epsilon)} d\epsilon,$$

where $C > 0$ is an absolute constant.

Corollary (Supremum of increments). Under the assumptions of the above Theorem,

$$\mathbb{E} \sup_{s, t \in T} |X_t - X_s| \leq 2CK \int_0^{\text{diam}(T, d)} \sqrt{\log \mathcal{N}(T, d, \epsilon)} d\epsilon.$$

Theorem (Uniform LLN for Lipschitz functions). Let μ be any probability measure on $[0, 1]$, and let $X_1, \dots, X_n$ be i.i.d. with law μ . For $L > 0$, define the anchored Lipschitz class

$$\mathcal{F}_L := \{f : [0, 1] \rightarrow \mathbb{R} : f(0) = 0, \|f\|_{\text{Lip}} \leq L\}.$$

Then

$$\mathbb{E} \sup_{f \in \mathcal{F}_L} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] \right| \leq \frac{CL}{\sqrt{n}},$$

where $C > 0$ is an absolute constant.

Remark The anchoring condition $f(0) = 0$ is only for convenience. Indeed, for any constant c ,

$$\mu_n(f + c) - \mu(f + c) = \mu_n(f) - \mu(f),$$

so adding constants does not affect the empirical process. Thus the same theorem can be read as a uniform law of large numbers over all Lipschitz functions modulo constants.

12. Empirical Process Theory

Empirical measure Define the empirical measure

$$\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{X_i},$$

where δ_x is the Dirac mass at x . Then for any measurable f ,

$$\mu_n(f) = \int f d\mu_n = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

So the approximation error is

$$\mu_n(f) - \mu(f).$$

Definition (Empirical process). Given a class $\mathcal{F}$ of real-valued functions on Ω , define

$$Z_f := \mu_n(f) - \mu(f) = \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}f(X), \quad f \in \mathcal{F}.$$

The random family $(Z_f)_{f \in \mathcal{F}}$ is called the *empirical process* indexed by $\mathcal{F}$.

The quantity of interest is then

$$\sup_{f \in \mathcal{F}} |Z_f| = \sup_{f \in \mathcal{F}} |\mu_n(f) - \mu(f)|.$$

A bound that tends to zero with n is called a *uniform law of large numbers* over $\mathcal{F}$.

Theorem (Symmetrization). Let $\mathcal{F}$ be a class of measurable real-valued functions on Ω , and let $\varepsilon_1, \dots, \varepsilon_n$ be i.i.d. Rademacher random variables, independent of everything else. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - \mathbb{E}f(X)) \right| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|.$$

Definition (Rademacher complexity). Let $\mathcal{F}$ be a class of measurable real-valued functions on Ω . Given a sample $X_1, \dots, X_n$, define the empirical Rademacher complexity

$$\hat{\mathfrak{R}}_n(\mathcal{F}) := \mathbb{E}_\varepsilon \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|,$$

where the expectation is taken only over the Rademacher variables $\varepsilon_1, \dots, \varepsilon_n$.

The expected Rademacher complexity is

$$\mathfrak{R}_n(\mathcal{F}) := \mathbb{E}_{X_1, \dots, X_n} \hat{\mathfrak{R}}_n(\mathcal{F}).$$

Theorem above can now be rewritten as

$$\mathbb{E} \sup_{f \in \mathcal{F}} |\mu_n(f) - \mu(f)| \leq 2 \mathfrak{R}_n(\mathcal{F}).$$

So Rademacher complexity measures how large the empirical process can be.

Definition (Traces of Boolean functions). Let $\mathcal{H}$ be a class of Boolean functions on Ω . Given sample points $x_1, \dots, x_n \in \Omega$, define the set of *traces* of $\mathcal{H}$ on the sample by

$$\mathcal{H}|_{x_{1:n}} := \{(h(x_1), \dots, h(x_n)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^n.$$

So $\mathcal{H}|_{x_{1:n}}$ records all labelings that the class can realize on the sample.

Definition (Growth function). For a class $\mathcal{H}$ of Boolean functions, define its growth function by

$$\Pi_{\mathcal{H}}(n) := \sup_{x_1, \dots, x_n \in \Omega} |\mathcal{H}|_{x_{1:n}}|.$$

The growth function measures how many different binary labelings the class can realize on n points.

Definition (Shattering). Let $\mathcal{H}$ be a class of Boolean functions on Ω , and let $\Lambda = \{x_1, \dots, x_m\} \subseteq \Omega$ be finite. We say that $\mathcal{H}$ *shatters* Λ if every labeling of Λ is realized by some function in $\mathcal{H}$. Equivalently, we have

$$|\mathcal{H}|_{\Lambda}| = 2^m.$$

Theorem (Sauer–Shelah). Let $\mathcal{H}$ be a class of Boolean functions with VC dimension $d < \infty$. Then for every $n \geq d$,

$$\Pi_{\mathcal{H}}(n) \leq \sum_{k=0}^d \binom{n}{k} \leq \left(\frac{en}{d}\right)^d.$$

Definition (VC dimension). The *VC dimension* of $\mathcal{H}$, denoted $\text{VC}(\mathcal{H})$, is the largest integer d such that some subset of Ω of size d is shattered by $\mathcal{H}$. If arbitrarily large finite sets are shattered, we write $\text{VC}(\mathcal{H}) = \infty$.

Theorem (VC bound for the uniform empirical error). Let $\mathcal{H}$ be a class of Boolean functions on Ω , and assume

$$d := \text{VC}(\mathcal{H}) < \infty.$$

Let $X_1, \dots, X_n$ be i.i.d. with law μ , and assume $n \geq d$. Then

$$\mathbb{E} \sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^n h(X_i) - \mathbb{E}h(X) \right| \leq C \sqrt{\frac{d \log(en/d)}{n}},$$

and in particular

$$\mathbb{E} \sup_{h \in \mathcal{H}} \left| \frac{1}{n} \sum_{i=1}^n h(X_i) - \mathbb{E}h(X) \right| \lesssim \sqrt{\frac{d \log n}{n}}.$$

13. Statistical Learning Theory

Supervised learning Let (X, Y) be a random pair taking values in $\mathcal{X} \times \mathcal{Y}$, where $\mathcal{X}$ is the feature space and $\mathcal{Y}$ is the label space. We observe training data

$$(X_1, Y_1), \dots, (X_n, Y_n),$$

which are i.i.d. copies of (X, Y) .

A **hypothesis** is a measurable function $h : \mathcal{X} \rightarrow \mathcal{Y}$ (or into some prediction space). We do not search over all possible functions, but only over a prescribed hypothesis class $\mathcal{H}$.

(Population) risk For a hypothesis $h \in \mathcal{H}$, define the *risk*

$$R(h) := \mathbb{E}[\ell(h(X), Y)].$$

This is the population prediction error. Since the distribution of (X, Y) is unknown, $R(h)$ is not directly computable.

The observable proxy is the **empirical risk**

$$R_n(h) := \frac{1}{n} \sum_{i=1}^n \ell(h(X_i), Y_i).$$

The population-optimal hypothesis in the class is

$$h^* \in \arg \min_{h \in \mathcal{H}} R(h),$$

while the empirical risk minimizer is

$$\hat{h}_n \in \arg \min_{h \in \mathcal{H}} R_n(h).$$

This is the **empirical risk minimization principle**: replace the unknown population risk by the empirical one, and minimize that instead.

Measures of errors. The quantity $R(h^*)$ is the best risk achievable within the class $\mathcal{H}$. It reflects the *approximation* quality of the class. The additional error

$$R(\hat{h}_n) - R(h^*)$$

is the **excess risk**: the price we pay for learning from finitely many samples.

Thus

$$R(\hat{h}_n) = R(h^*) + (R(\hat{h}_n) - R(h^*)),$$

which separates the total prediction error into an approximation term and a statistical term.

Excess risk bound. For any hypothesis class $\mathcal{H}$,

$$R(\hat{h}_n) - R(h^*) \leq 2 \sup_{h \in \mathcal{H}} |R_n(h) - R(h)|.$$

Generalization bound (fixed h). Since $\ell \in [0, 1]$, by Hoeffding's inequality,

$$\mathbb{P}\{|R_n(h) - R(h)| \geq t\} \leq 2 \exp(-2nt^2).$$

Theorem (VC generalization bound). Let $\mathcal{H}$ be a class of Boolean hypotheses with finite VC dimension

$$d := \text{VC}(\mathcal{H}) < \infty.$$

Assume $n \geq d$. Then

$$\mathbb{E}[R(\hat{h}_n) - R(h^*)] \leq C \sqrt{\frac{d \log(en/d)}{n}},$$

where $C > 0$ is an absolute constant. In particular,

$$\mathbb{E}[R(\hat{h}_n) - R(h^*)] \lesssim \sqrt{\frac{d \log n}{n}}.$$

14. Good to know

1. $\Gamma(z+1) \leq (C_0 z)^z$
2. $\delta^2 = \min\{t, t^2\} \iff t = \max\{\delta, \delta^2\} \leq \delta + \delta^2$
3. $\int_0^\infty t^{p-1} e^{-\alpha t^2} dt = \frac{1}{2} \alpha^{-p/2} \Gamma(p/2)$
4. $\int_0^\infty t^{p-1} e^{-\alpha t} dt = \alpha^{-p} \Gamma(p)$